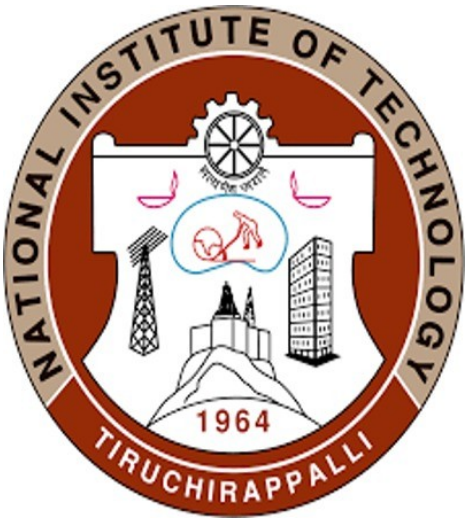
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MOSFET-Based 2-Way Sallen-Key Active Audio Filter

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1. INTRODUCTION

An audio signal is a voltage that varies in the same way as the sound it represents. Human hearing covers roughly 20 Hz to 20 kHz; the low end of this range is bass and the high end is treble. Music is a combination of many such frequencies at the same time.

A single loudspeaker driver cannot reproduce this whole range well. A woofer uses a large cone that can move enough air for bass, but its mass prevents it from following high frequencies. A tweeter uses a small, light dome suited to treble, but large low-frequency signals can overdrive and damage its voice coil. Multi-driver speakers therefore use a **crossover**, which splits the signal so that each driver receives only its own band. In a 2-way crossover, a low-pass filter (LPF) feeds the woofer and a high-pass filter (HPF) feeds the tweeter. The frequency at which the two filters pass equal signal is the **crossover frequency**, f_c. We chose $f_c \approx 1$ kHz.

In this project we built a 2-way active crossover, which filters the small line-level signal before the power amplifiers. Each channel is a second-order Sallen-Key filter, which uses two resistors, two capacitors and a unity-gain buffer. The buffer is usually an op-amp; here it is replaced by a single 2N7000 MOSFET connected as a source follower, running from a single +9 V supply.

2. PROBLEM STATEMENT

To design and build a 2-way active audio crossover with $f_c \approx 1$ kHz using second-order Sallen-Key low-pass and high-pass filters, in which the buffer is a 2N7000 MOSFET source follower instead of an op-amp, operating from a single +9 V supply.

Using a MOSFET instead of an op-amp introduces the following problems:

1. The source follower gain is less than one (about 0.94), while the standard Sallen-Key design assumes a gain of exactly one.
2. With a gain below one, the quality factor Q drops, so the component values must be chosen to keep Q close to 0.707 (Butterworth).
3. The MOSFET conducts only when $V_{GS} > V_t$, so with a single supply the gate must be biased at a DC level near mid-supply without loading the filter.

3. CIRCUIT AND CALCULATIONS

3.1 Circuit Diagram

The circuit was drawn and simulated in LTspice. Figure 1(a) shows Channel 1, the low-pass filter (woofer output), and Figure 1(b) shows Channel 2, the high-pass filter (tweeter output). The audio input (IN) feeds both channels in parallel, and each channel has its own 2N7000 source follower. Both channels are powered from a single +9 V supply (VCC), decoupled by 100 µF and 100 nF capacitors.

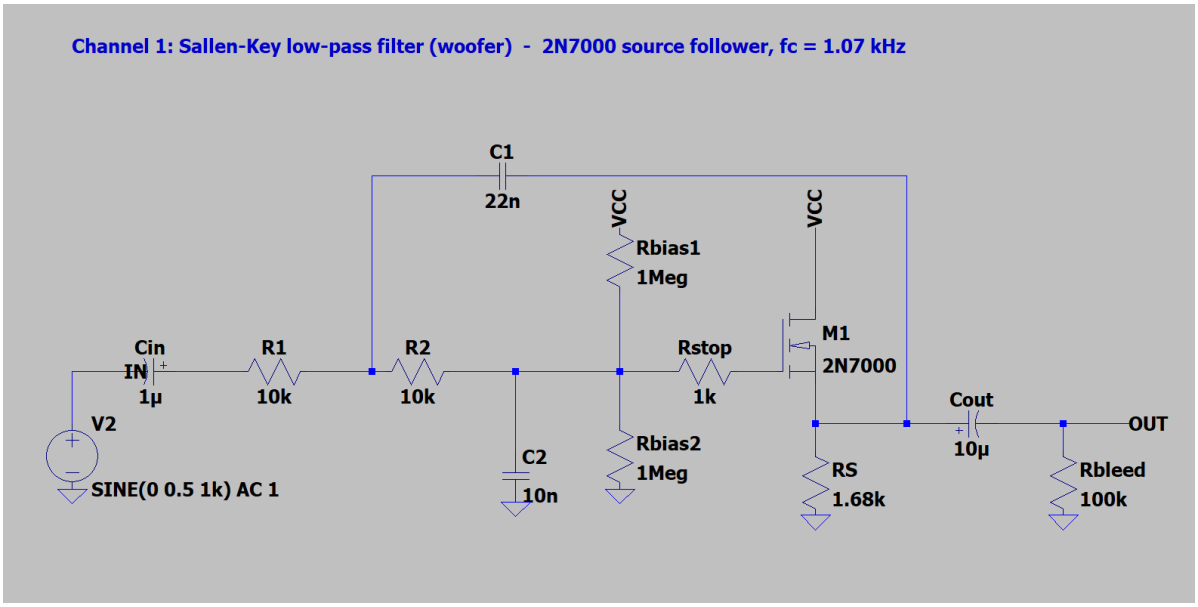


Figure 1(a): Channel 1, Sallen-Key low-pass filter (woofer)

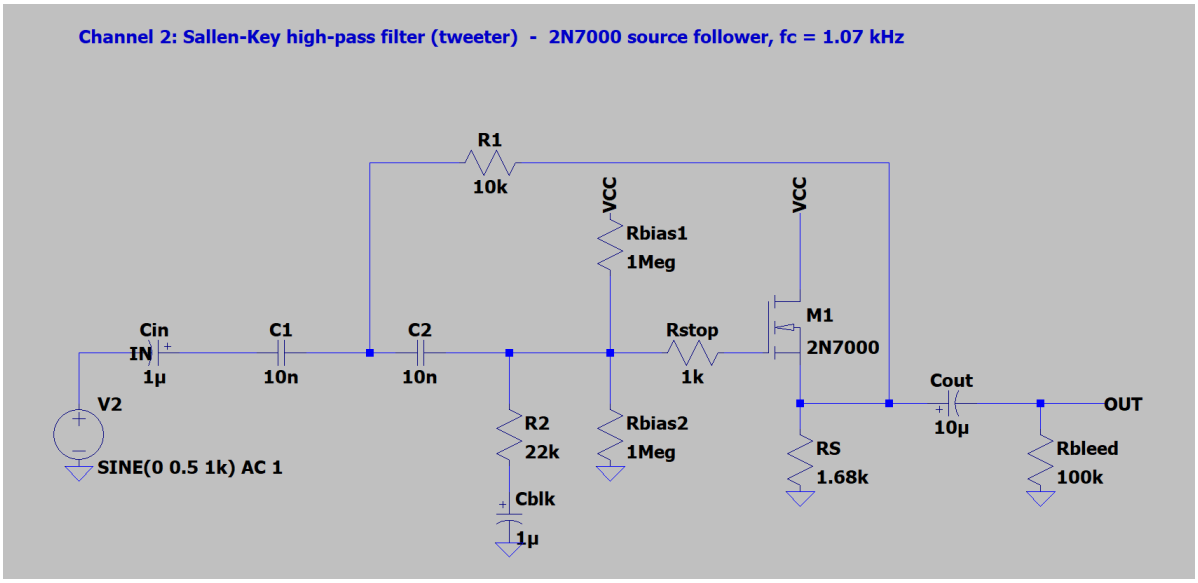


Figure 1(b): Channel 2, Sallen-Key high-pass filter (tweeter)

Table 1: Component values

Component	Value	Purpose
Q1, Q2	2N7000	Source-follower buffer
R1, R2 (LPF)	10 kΩ each	Series resistors
C1, C2 (LPF)	22 nF, 10 nF	Feedback and shunt capacitors
C1, C2 (HPF)	10 nF each	Series capacitors
R1, R2 (HPF)	10 kΩ, 22 kΩ	Feedback and shunt resistors
C_{blk}	1 µF	Blocks DC through R2 (HPF)
$R_{bias1} - R_{bias4}$	1 MΩ each	Gate bias dividers
R_{stop1}, R_{stop2}	1 kΩ	Gate stopper resistors
R_{S1}, R_{S2}	1.68 kΩ	Source resistors
C_{in}, C_{out}	1 µF, 10 µF	Input and output coupling
R_{bleed1}, R_{bleed2}	100 kΩ	Hold outputs at 0 V DC
C_{dec1}, C_{dec2}	100 µF, 100 nF	Supply decoupling

3.2 DC Biasing

The gate is biased at mid-supply by two equal 1 MΩ resistors:

$$V_G = V_{DD}\,\frac{R_{bias2}}{R_{bias1}+R_{bias2}} = 9 \times \frac{1}{2} = 4.5\ \text{V}$$

Large values are used so that the AC resistance at the gate, 1 MΩ ∥ 1 MΩ = 500 kΩ, is much larger than the 10 kΩ filter resistors. Taking $V_{GS} \approx 2$ V:

$$V_S = V_G - V_{GS} \approx 2.5\ \text{V}, \qquad I_D = \frac{V_S}{R_S} = \frac{2.5}{1.68 \times 10^3} \approx 1.49\ \text{mA}$$

$V_{DS} \approx 9 - 2.5 = 6.5$ V, which is much larger than $V_{GS} - V_t$, so the MOSFET operates in saturation.

3.3 Source Follower Gain

At this drain current the transconductance of the 2N7000 is about $g_m \approx 9$ mS, so with $R_S = 1.68$ kΩ:

$$A_v = \frac{g_m R_S}{1 + g_m R_S} = \frac{15.1}{16.1} \approx 0.94$$

The output resistance of the source follower is about 100 Ω, low enough to drive the 10 kΩ filter components. The gain K = 0.94 is used for the filter calculations below.

3.4 Low-Pass Filter

In the low-pass filter, C1 (22 nF) is the feedback capacitor and C2 (10 nF) is the capacitor to ground. The cutoff frequency is:

$$f_c = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}}$$

With $R_1 = R_2 = 10$ kΩ, $C_1 = 22$ nF and $C_2 = 10$ nF:

$$f_c = \frac{1}{2\pi\sqrt{(10^4)(10^4)(22\times10^{-9})(10\times10^{-9})}} \approx 1073 \text{ Hz}$$

For equal resistors and an ideal buffer, Q depends only on the capacitor ratio:

$$Q = \frac{1}{2}\sqrt{\frac{C_1}{C_2}} = \frac{1}{2}\sqrt{\frac{22}{10}} = 0.742 \quad (K = 1)$$

The MOSFET gain of 0.94 lowers this to about 0.70, which is close to the Butterworth value of 0.707. This is why a 2.2 : 1 capacitor ratio was used instead of equal capacitors (which would give Q = 0.5).

3.5 High-Pass Filter

The high-pass filter is obtained by interchanging the resistors and capacitors. R1 (10 kΩ) is the feedback resistor and R2 (22 kΩ) is the resistor to ground. Using the same formula with $C_1 = C_2 = 10$ nF, $f_c \approx 1073$ Hz. For equal capacitors and an ideal buffer:

$$Q = \frac{1}{2}\sqrt{\frac{R_2}{R_1}} = \frac{1}{2}\sqrt{\frac{22}{10}} = 0.742 \quad (K = 1)$$

With the MOSFET gain of 0.94, Q ≈ 0.70, the same as the low-pass filter. C_{blk} (1 µF) is placed in series with R2. Without it, R2 would pull the gate bias down to about 0.2 V and the MOSFET would turn off. Its corner frequency is well below the audio band:

$$f_{blk} = \frac{1}{2\pi(22\text{ k}\Omega)(1\ \mu\text{F})} \approx 7 \text{ Hz}$$

3.6 Other Components

The input capacitor keeps the 4.5 V gate bias away from the signal source, and the output capacitor with the 100 kΩ bleed resistor removes the DC level at the source so that the output sits at 0 V. The 1 kΩ gate stopper damps high-frequency oscillation, and the 100 µF and 100 nF capacitors decouple the supply rail.

3.7 Theoretical Response

Figure 2 shows the calculated magnitude and phase response of both channels. The two curves cross at 1.07 kHz, where each output is about 3.7 dB below the input and the two outputs are 180° apart.

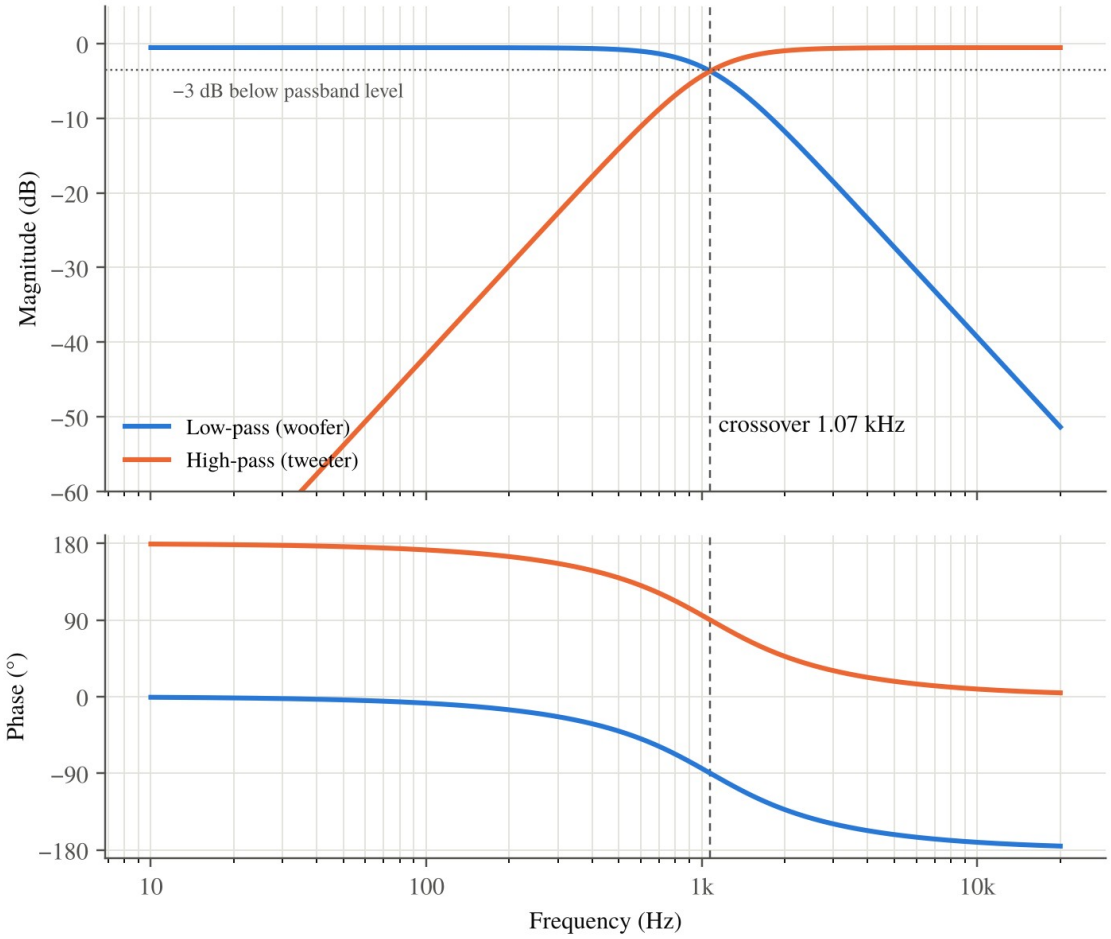


Figure 2: Calculated magnitude and phase response of the low-pass and high-pass channels

4. RESULTS AND DISCUSSION

4.1 Setup

The circuit was built on a breadboard and powered from a 9 V battery. A sine wave from a function generator was applied to both channels. DC voltages were measured with an SEW DT830D multimeter (20 V DC range) and waveforms were observed on a GW Instek GDS-1102B DSO.

4.2 DC Measurements

Table 2: Measured DC voltages (no input signal)

Measurement	Expected	Measured
Gate to GND	4.5 V	3.2 V
+9 V rail to gate	4.5 V	3.2 V
Source to GND	2.2 – 2.7 V	2.75 V

The gate reading is lower than expected because the multimeter has an input resistance of about 1 MΩ. When it is connected, it appears in parallel with one of the 1 MΩ bias resistors (giving 0.5 MΩ), so the meter reads about one third of the supply instead of one half. The same applies when measuring from the +9 V rail to the gate. Since both readings were equal (3.2 V), the divider is symmetric and the supply is about 3 × 3.2 = 9.6 V, so the actual gate voltage is about 4.8 V. The source is a low-impedance node, so its reading is not affected by the meter. From the measurements:

$$V_{GS} \approx 4.8 - 2.75 = 2.05 \text{ V}, \qquad I_D = \frac{2.75}{1.68 \times 10^3} \approx 1.64 \text{ mA}$$

This is close to the design value of 1.49 mA, and $V_{DS} \approx 6.9$ V confirms that the MOSFET is in saturation.

4.3 Frequency Response

Both channels were tested with a 1 V_{pp} sine input. Table 3 lists the measured output and gain (V_{out}/V_{in}) along with the calculated gain. Readings marked * are close to the noise floor of the setup (about 30 – 100 mV) and do not represent the true filter output.

Table 3: Measured frequency response (V_{in} = 1 V_{pp})

f (Hz)	LPF V_{pp} (mV)	LPF gain	HPF V_{pp} (mV)	HPF gain	Calc. LPF	Calc. HPF
50	960	0.96	–	–	0.94	0.002
100	950	0.95	40*	0.04*	0.94	0.01
200	944	0.94	100*	0.10*	0.94	0.03
500	888	0.89	360	0.36	0.91	0.20
1000	690	0.69	780	0.78	0.70	0.61
1070	640	0.64	820	0.82	0.65	0.65
2000	320	0.32	952	0.95	0.26	0.90
5000	100*	0.10*	992	0.99	0.04	0.94
10000	30*	0.03*	1000	1.00	0.01	0.94

The LPF output stays near 0.95 V up to 200 Hz and falls to 0.64 V at 1.07 kHz, while the HPF output rises from 0.36 V at 500 Hz to about 1 V above 5 kHz. Taking the −3 dB point as 0.707 of the passband value, the measured cutoff frequency is about 1.01 kHz for the LPF and about 0.86 kHz for the HPF. The two outputs are equal at about **0.9 kHz**, which is the measured crossover frequency. Figure 3 compares the measured gain with the calculated response.

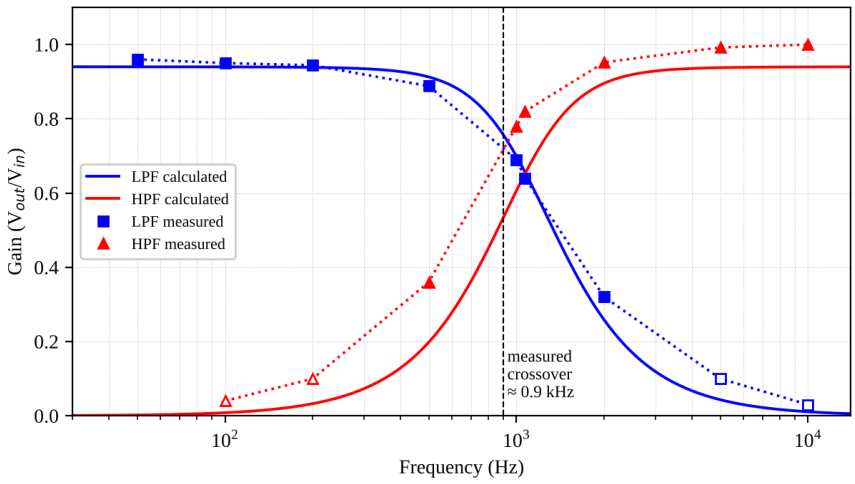


Figure 3: Measured and calculated gain of both channels (hollow markers: readings at the noise floor)

4.4 Discussion

- The measured passband gain (0.95 – 1.0) agrees with the calculated source follower gain of 0.94 within the accuracy of the function generator amplitude setting.
- The measured cutoff frequencies are lower than the calculated 1.07 kHz by about 5% (LPF) and 20% (HPF). Since $f_c \propto 1/\sqrt{(R_1R_2C_1C_2)}$, this is mainly due to the tolerance of the 10 nF and 22 nF ceramic capacitors (typically ±10 – 20%).
- The LPF gain at 2 kHz (0.32) is higher than calculated (0.26), indicating a slightly lower Q (about 0.6) than designed. This is caused by the actual C_1 : C_2 ratio being less than 2.2 : 1.

- Since the HPF cutoff is below the LPF cutoff, the two bands overlap slightly and the outputs cross at about −2.7 dB instead of −3 dB.
- R_S does not appear in the f_c expression. It only sets the bias current and the follower gain.
- The LPF and HPF outputs are 180° apart at f_c, so one of the drivers should be connected with reversed polarity to avoid a dip in the combined response.
- The outputs are line-level and must be followed by a power amplifier to drive speakers.

5. APPLICATIONS

1. **Bi-amplified loudspeakers and studio monitors**, where the woofer and tweeter each have their own power amplifier, avoiding the losses of passive crossover inductors.
2. **Subwoofer bass management** in home theatre systems, using a low-pass filter (typically 80 – 120 Hz) for the subwoofer and a high-pass filter for the satellite speakers.
3. **Musical instrument electronics** such as guitar effects, equalisers and synthesisers, for tone shaping and band splitting.
4. **Anti-aliasing filters** placed before analog-to-digital converters to remove frequencies above half the sampling rate.
5. **Low-cost signal conditioning** where an op-amp is not available or the supply voltage is higher than an op-amp can handle.

6. FUTURE SCOPE

- **3-way crossover:** adding a band-pass channel for a mid-range speaker.
- **Linkwitz-Riley response:** cascading two Butterworth sections per channel, so the outputs are in phase at f_c and sum to a flat response.
- **Adjustable crossover frequency:** using dual-gang potentiometers for R1 and R2.
- **Improved buffer:** replacing R_S with a constant-current source to bring the gain closer to one.
- **PCB layout:** a PCB with a ground plane to reduce the noise seen on the DSO, and closer-tolerance (C0G or film) capacitors to bring f_c closer to the design value.

7. REFERENCES

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